CHAROTAR UNIVERSITY OF SCIENCE&TECHNOLOGY

__II__ Semester of Bachelor of Optometry Examination April 2018

BOP205 Geometrical Optics II

Date: 19/04/2018 Day: Thursday Time: 10.00 am To 10.30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I (A)	**Choose the correct answer for the following questions.**		**10**
1.	**In case of compound myopic astigmatism light rays from both the meridians focuses_________**		
	a. On the retina	b. In front of retina	
	c. Behind the retina	d. None of the above	
2.	**_________ test is based on preferential looking principle**		
	a. LogMAR chart	b. Snellen chart	
	c. Jaeger chart	d. Lea paddle	
3.	**Refractive power of +10D Biconvex lens has focal length of_____**		
	a. 10 cm	b. 20cm	
	c. 15cm	d. 5cm	
4.	**Hyperopic refractive error is corrected by______**		
	a. Plus lens	b. Minus lens	
	c. Prism	d. None of the above	

5.	**Light rays coming from the infinity has____ vengeance**		
	a. Positive	b. Negative	
	c. Zero	d. None of the above	
6.	**In Aphakic eye the posterior focal point is about____ behind the eyeball.**		
	a. 25mm	b. 30mm	
	c. 31mm	d. 40mm	
7.	**Diopter is defined as inverse of _____ in meters**		
	a. Focal length	b. Radius of curvature	
	c. Lens diameter	d. Refractive index	
8.	**Stimulus for accommodation is ___**		
	a. Blur retinal image	b. Chromatic aberration	
	c. Proximity of an object	d. All of the above	
9.	**Visual acuity depends on following factors except**		
	a. Distance between object and eye	b. Colors of object	
	c. Time of exposure	d. Contrast of object with its background	
10.	**Reduced schematic eye is having ____number of refracting surface**		
	a. 1	b. 2	
	c. 3	d. 4	

Q - I (B)	**Match the following column "A" with Column "B".**	**5**
Column A	**Answer**	**Column B**
1. Cardiff acuity card		a. Gullstrand
2. Schematic eye		b. Pediatric patient
3. Long sightedness		c. Absence of lens
4. Aphakia		d. Hyperopia
5. Stops		e. Spherical aberration

Q - I (C)	Select whether following statement is true or false.	5
		True/False
1.	Presbyopia is a refractive error	
2.	In case of thin lens, lens thickness is negligible compare to the radius of curvature of lens surface	
3.	Punctum remotum means nearest point at which things can be seen clearly	
4.	Ability to identify smallest target possible is known as a resolving acuity	
5.	Coma is on the axis aberration	

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Date: 19/04/2018 Day: Thursday Time: 10.30 am to 01.00 pm Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I		
Q - II	**Answer the following questions as directed. (Any ten)**	**20**
1.	Define Presbyopia	
2.	Enlist use of Schematic eye	
3.	Explain Vernier acuity	
4.	What is Detection acuity?	
5.	Define Visual axis and optical axis of eye	
6.	Define vertex distance.	
7.	Define Range of accommodation and Amplitude of accommodation	
8.	Explain effective power.	
9.	Write down Hofstetter's formula with its use.	
10.	Define Astigmatism.	
11.	Define pseudophakia.	
SECTION – II		
Q-III	**Answer the following questions as directed (Any four)**	**20**
1.	Describe in detail about optics of Aphakia.	
2.	Explain spectacle magnification and relative spectacle magnification.	
3.	Write a note on mechanism and reaction time for accommodation	
4.	Explain two methods to determine the presbyopic correction.	
5.	Write down signs of pseudophakia	
Q-IV	**Answer the following questions as directed**	**10**
1.	Enlist the name of the test used to assess visual acuity in pediatric patient and explain any two in detail. **Or** Classify astigmatism and draw diagram for each type of astigmatism.	